Avantika Dhananjay Patil

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.30
DAV Public School	Class XII (CBSE)	2023	97.60%
DAV Public School	Class X (CBSE)	2021	97.40%

SCHOLASTIC ACHIEVEMENTS

- Selected for the Sakura Science Cultural Exchange Program, **Japan** (2025).
- **City Topper** in CBSE XII with 97.6% (Centinium in Physics).

WORK EXPERIENCE

American Express - Data Analyst Intern, Control Management

Tools: Generative AI, GPTs

- Designed **AI-assisted control-testing workflows** using **10+** RCSA knowledge bases to resolve tester queries.
- Automated risk-metric anomaly screening across **150-page reports**, reducing manual effort by approx **5 hours**.
- Built **AI-driven control QA** to score evaluations and generate Excel reports with gaps and remediation reasons.

Research Internship at IUCAA - Data Analysis and Model Fitting. GitHub

Tools: Python, NumPy, SciPy, Cobaya, MCMC

- Applied **MCMC methods** to fit models and extract posterior parameter distributions from **-decay** datasets.
- Constructed and analyzed **covariance and correlation matrices** to understand model sensitivity.
- Performed **Bayesian model comparison** (AICc, 2, p-values) to select optimal models.

RESEARCH EXPERIENCE

Belle II - Reconstruction of Mesons Leading to + Final States

Tools: C++, ROOT/RooFit, basf2

- Reconstructed decay and corrected low-energy photon bias, improving 0 and mass resolution by **8%**.
- Diagnosed **generator-level mis-modelling issue** by reconstructing a0(980), identifying MC simulation inaccuracies.
- Reconstructed D+s + and D+ +; performed mass fits and **momentum-dependent data-MC comparisons.**

Study of Rare Decays at Belle II

Tools: basf2, ROOT, RooFit, Python

- Performed statistical signal extraction to established **90%** C.L. upper limits at **317** (ee) and **44** ( ) events.
- Developed a **reconstruction and analysis pipeline** for rare meson decays using collision data and **Monte Carlo simulations**.
- Optimized background rejection through **photon-conversion suppression** and **Bremsstrahlung recovery** techniques.

PROJECTS

Lucy - Autonomous Software Engineering Copilot AI Agent. GitHub

Tools: Strands Agents, Langsmith, Python

- Built a **Strands AI Agent** runtime supporting planning, tool orchestration, and repository-aware context management.
- Implemented resilient execution through check-pointing, retry strategies, failure recovery, and graceful tool degradation.
- Designed modular abstractions separating reasoning, execution, and tools for extensible agent workflows.
- Engineered interactive terminal workflows with streaming updates, **approval gates**, and structured observability.

Decoding Airfare Volatility and Dynamic Pricing using Machine Learning. GitHub Drive

Tools: Python, XGBoost, LightGBM, CatBoost, PySpark

- Analyzed a **31 GB Expedia flight-pricing** dataset containing nearly **6 million** records using PySpark pipelines.
- Implemented the **Extended Isolation Forest** to statistically remove data outliers, improving model fit by nearly **10 %.**
- Developed and compared Random Forest, XGBoost, LightGBM and CatBoost models for ticket-price forecasting.
- Conducted OLS Regression to prove a **9.4%** airfare hike, discovering a **4-6 week** prior window for optimal fares.

Earthquake Forecasting using Statistical Seismology and Machine Learning. Github

Tools: Python, XGBoost, NumPy, SciPy, Pandas

- Analyzed **30 years** of California seismic data to study earthquake occurrence patterns and tectonic stress evolution.
- Trained an XGBoost model achieving **ROC-AUC of 0.86** and **Recall of 0.73** for moderate-to-large earthquakes.
- Estimated **Gutenberg-Richter** parameters using Max Likelihood Estimation and performed rolling b-value analysis.
- Engineered spatio-temporal features including event rates, spatial variance and **cumulative Benioff strain.**

SKILLS

Programming: Python, C, C++, SQL, Git

Machine Learning: Scikit-Learn, XGBoost, CatBoost, LightGBM

Gen-AI: Strands Agents, Langsmith.

Tools: NumPy, Pandas, SciPy, Matplotlib, ROOT/RooFit, basf2, Cobaya

RELEVANT COURSES

- **Computer Science**: Data Science and Analysis, ML in Physical Sciences , DSA, Introduction to Programming (C), Introduction to Artificial Intelligence.
- **Mathematics**: Probability and Random Variables, Advanced Mathematical Physics, Linear Algebra, Differential Equations, Calculus I & II.
- **Physics**: Special Relativity, Computational Physics, Classical Mechanics, Statistical Mechanics, Quantum Mechanics, Electrodynamics, Solid State Physics, Atomic and Molecular Physics.

EXTRACURRICULARS / POSITIONS OF RESPONSIBILITY

- **Finance Head** - Prayas (Social Club), IIT Hyderabad (2024-2025)
- **Core Member** - Kludge (Cybersecurity & Networking Club) (2024-2025)
- **Core Member** - Rang De Manch (Dramatics Club) (2024-2025)